The Eight Jobs: A Mirror for Civilisations

By Kari McKern

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I am a retired public servant and library IT specialist from Sydney, not an academic. For forty years I carried a question that had no department: why does every empire leave the same skeleton in the dust? This essay is about the answer my machines and I have built, what it has seen so far, and what it cost me to find out. Every number in it comes from the open record at neuralnations.org and can be checked. I would rather be checked than believed.

The Map Beneath the Weather

The political theatre, the thunder and lightning of left and right, of tyrant and liberator, is just weather. The real climate is the load-bearing structure underneath, and it was sitting in the open the whole time, glinting in the dust like a shiny penny on a crowded footpath. Millions of boots step over it every day because everyone is looking up at the billboards.

Here is the penny. A lone wolf can bring down a deer. It cannot touch a bison, and a bison can kill one with a single swing of its head. A pack can take the bison. The pack does not make any wolf stronger; it makes prey reachable that no wolf can reach alone. Scale that up and the same arithmetic holds for a family, a firm, a nation. Any group that keeps cooperating over time has exactly eight jobs to do. I gave the jobs names with some myth in them, because names that carry pictures are easier to remember than departments: **Helm** decides, **Shield** defends, **Lore** means, **Stewards** allocates, **Craft** makes, **Flow** trades, **Hands** works, and **Archive** remembers.

Every year, for every society in the corpus, the instrument takes thirty-two readings: each of the eight functions scored on coherence, capacity, stress and abstraction. Those readings, published in full, produce eight node values and twenty-eight bonds. The dataset behind this essay, the unified panel, holds forty-eight societies and 5,425 scored society-years, some of them running back two centuries and more. The arithmetic is frozen and public, and it verifies to within four parts in a quadrillion.

A word about what the instrument refuses to do, because it is the protection that matters most. Coercion is not coherence. A regime that suppresses error reports and forces compliance may look orderly, and it is scoring badly on exactly the dimension that counts, because under disturbance the information and the correction never travel. The audit question is not: is everyone obeying? It is: under disturbance, are critical information and error correction actually transmitted? A society can be obedient and incoherent at the same time. A great many have been.

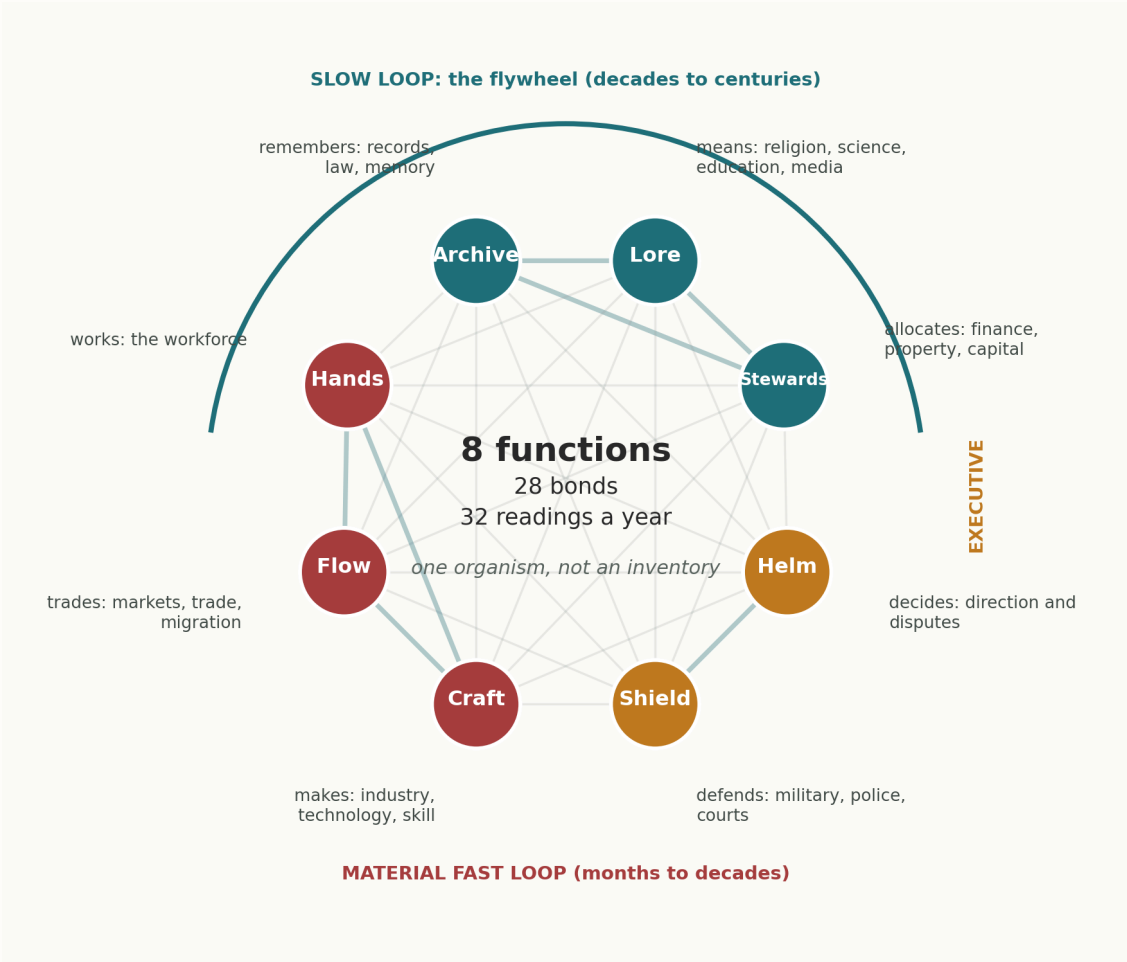


Figure 1. The eight functions and the twenty-eight bonds between them. Three layers, two clocks. A schematic, not a reading.

The Shape Between the Parts

The most important idea in the model sits between the functions, not inside them. Two hospitals can have identical staff, equipment and budgets and differ utterly in what they can do, because in one the theatre team, the blood bank and the administrator talk to each other, and in the other they file reports at each other. Nothing in the inventory shows the difference. The bonds do. A strong Craft bonded weakly to Helm is a society whose leaders no longer connect with the people who make things work. A Lore that has drifted away from Archive is a society telling itself stories its own records do not support.

The arithmetic of a bond is the arithmetic of a household with one person in crisis: the others do not stop loving them, they stop being able to coordinate with them, and that is a different and more dangerous thing.

When you score responsiveness across the whole panel, an anatomy falls out (Figure 4). Some functions barely move when the rest of the system moves: Lore responds at about three-quarters of the system average, an inelastic anchor that holds meaning steady while everything else swings.

Others overshoot: Flow and Helm amplify, responding at more than one and a quarter and nearly one and a fifth respectively. I published this ordering months ago; recomputing it on the corpus behind this essay reproduces it almost exactly. It is real structure, not a story I am telling.

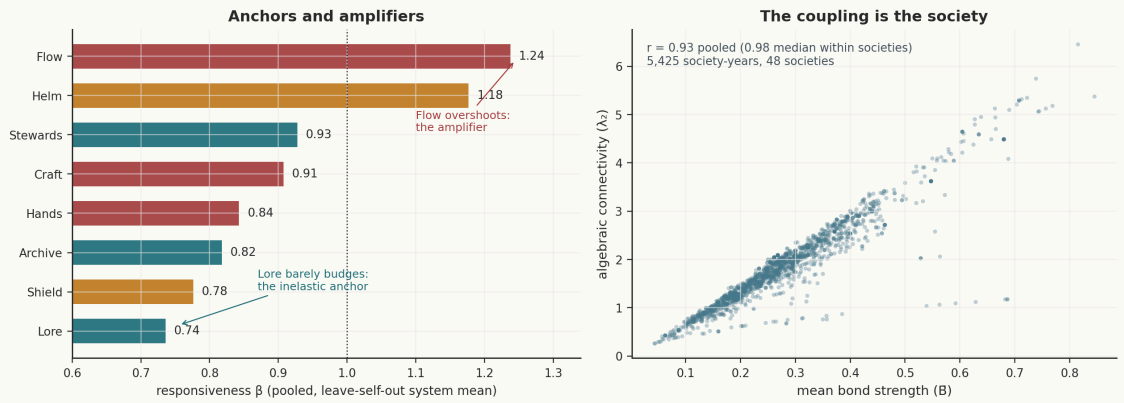


Figure 4. Left: responsiveness of each function to the rest of the system (pooled β on the leave-self-out system mean). Right: mean bond strength against algebraic connectivity across 5,425 society-years. Computed from the unified panel.

And the coupling itself is measurable. Mean bond strength and the network's algebraic connectivity track each other at r = 0.93 pooled across all society-years, 0.98 within societies. I report the pooled figure because it is the honest one; the prettier number is the within-society median. Roughly seven tenths of all year-to-year motion across the panel is one shared movement, one synchronised slide. A society is not eight instruments. It is one chord.

Two Clocks

There are two clocks in any society, and they run at very different speeds. The fast loop of budgets, elections, harvests and news cycles turns over in months to decades. The slow loop of the archive, the schools, the constitutional settlement and the stories a people tells about itself turns over in decades to centuries. The slow loop is the flywheel. It stores momentum, it smooths the shocks the fast loop cannot absorb, and it is ruinously expensive to restart once it has stopped. A bad government is a bad year. A lost archive is a lost century.

I call the formal version of the flywheel the Library Attractor: the bonding strength of the memory and meaning functions, measured against the stress on working hands. Think of a floodplain. A river with a floodplain can take a wet year without destroying the towns. A river without one destroys the towns and then looks calm.

Cambodia is the cautionary tale (Figure 2, left). In 1975 the Khmer Rouge took over, and the instrument's library floor collapsed to its nadir, a reading of 0.017 where healthy societies typically sit several times higher. The freeze ran for four years, and you can watch the whole system flatline with it.

That is what a thermodynamic freeze looks like from outside: coordination ceasing to generate positive value at all. The reading is not a metaphor. It is thirty-two scores a year, cold as a weather station.

The sharpest test I have written into the pre-registered program is the Amnesia Test: at the bottom of a catastrophe, does a society still remember what it is for? If the Library Attractor is real, amnesiac societies should struggle to restart even when their material capacity survives. If recovery tracks only material capacity, the attractor is an elegant fiction and must go. That is the one I would most like someone else to run.

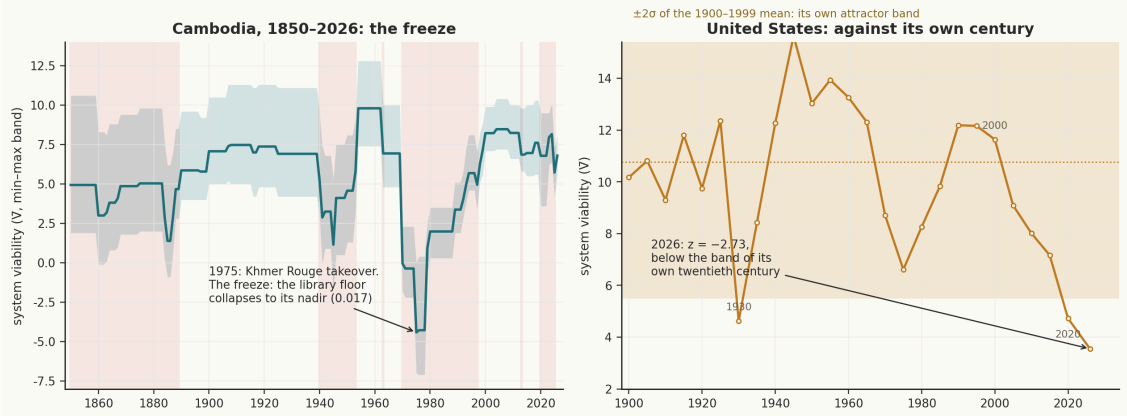


Figure 2. Left: Cambodia, 1850–2026, system viability with crisis years shaded; the 1975 nadir of the library floor is annotated. Right: the United States against its own 1900–1999 attractor band, sampled at five-year intervals; the annual ensemble series tells the same story. Computed from the unified panel, canonical operators.

What It Sees Now

I have no forecast to protect; the registered predictions are lodged and will be scored when the time comes, whatever they show. The reading is simply the reading. The United States in 2026 sits almost three of its own standard deviations below the band of its own twentieth century: z = −2.73, computed against its 1900–1999 mean (Figure 2, right). Russia sits inside its own band. China sits above its own mean, which is low, because China's twentieth century included collapses no American decade approached. Each country judged against itself: that is the fairest comparison the instrument allows. On the annual ensemble series, the weakest of the eight functions in America has been the Helm, the function that sets direction and settles disputes about it, for eleven consecutive years, a run with no precedent in the record except the years of Vietnam and Watergate. The instrument does not know which party holds the Helm. It knows the decision function has been the weakest link for over a decade. Whatever your politics, that is the reading.

And it is not only America. Between 2000 and 2025, in the thirty-eight societies of the panel with comparable years, twenty-two declined in mean system viability and twenty-five saw stress rise (Figure 3). The slide takes in the United States, France, Britain, Australia and New Zealand alongside

Venezuela and Syria. It is not a war effect, a sanctions effect, or a Western effect. Within a generation the panel gave back a large share of what the postwar decades built. The instrument measures strain, not schedule, and a reader who wants a date is asking the wrong instrument. But strain ignored has a schedule of its own, and it is rarely the one we would choose.

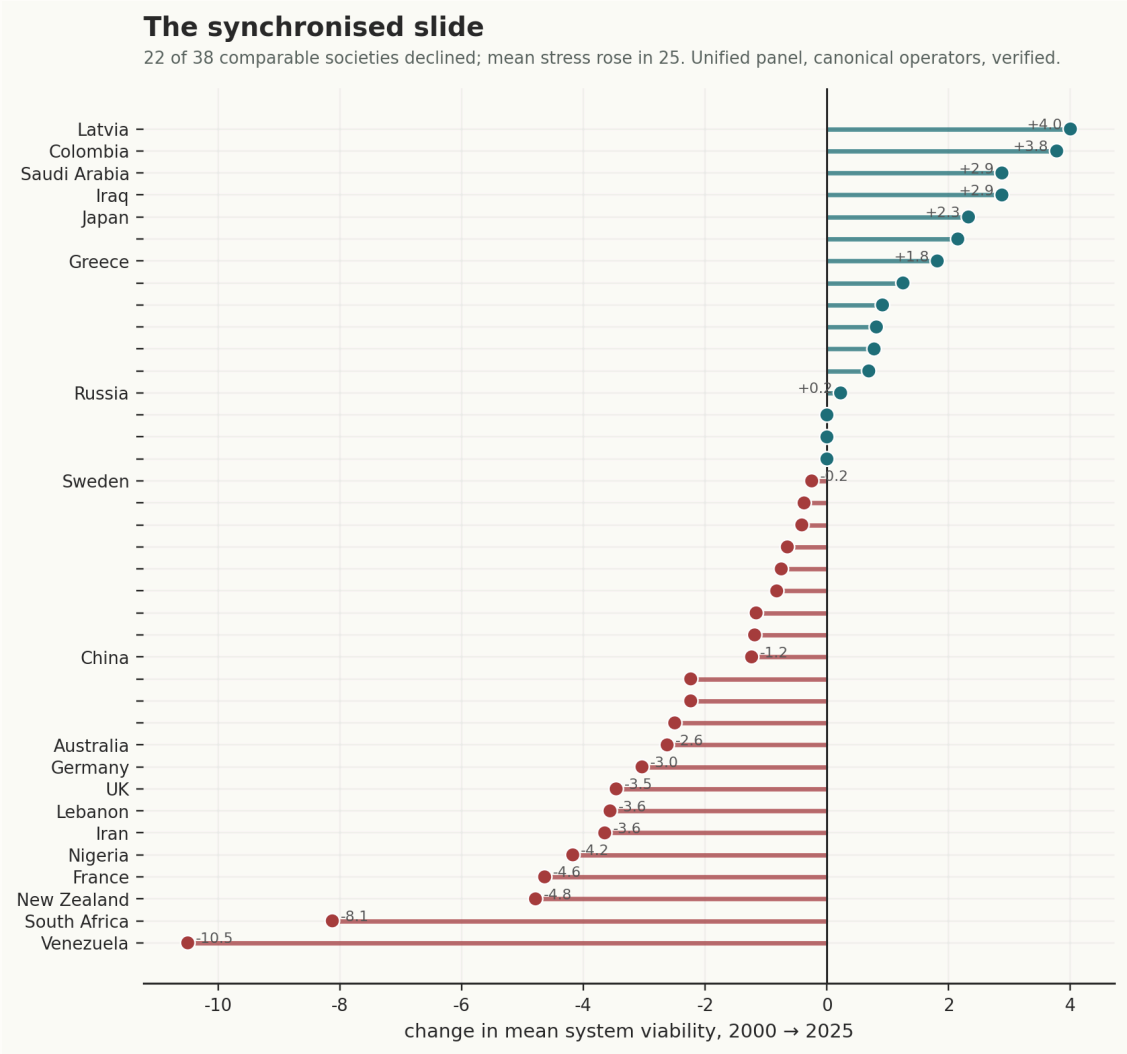


Figure 3. Change in mean system viability, 2000 to 2025, for the thirty-eight societies with comparable years. Twenty-two declined; mean stress rose in twenty-five. Computed from the unified panel.

I Would Rather Be Checked Than Believed

Now the heresy, and it is the part I care about most. This model has been wrong, publicly, in writing, more than once. My favourite composite operators turned out to be arithmetic talking to itself: when the machine audited one of my headline concordances, half of it was the equation conversing with its own reflection, and the honest remainder was a quarter of what I had claimed. A whole pre-registered programme of mine returned null last month, and the null is publishable, so it will be published. The idea that societies with certain shapes recover better failed outright, productively: displacement is real, prediction is not. The boldest single test of the network architecture, though, passed: the rank order of

the twenty-eight bonds, frozen and hashed before anyone opened the sealed holdout of twelve anonymised societies, matched at Spearman 0.702, p = 0.0000248. There are 40,320 ways to relabel eight nodes. Exactly one of them matched as well as the data did. The seven bonds that survived correction are the load-bearing relationships of any working organisation: direction and force, making and trading, memory and meaning, skill and labour. The theory did not name them in advance. The data produced them.

What is settled: three AI raters from three laboratories agree on blinded historical material at r = 0.89 to 0.93, and the instrument reproduces its own arithmetic exactly. What is open: independent human raters, who alone can separate shared reading from shared memory; whether seven or nine functions would serve better; and the registered forecasts, which will be scored when the time comes, whatever they show. Eight nodes are a working grammar, not a discovery of nature’s joints. The machines demanded evidence from each other and often forced me to abandon ideas I had grown fond of. That is what they are for. An exosuit that only ever agreed with you would be a corset.

Carl Sagan asked who speaks for Earth. The stranger thing has happened: a short-lived primate built a mirror large enough to read the state of its own cooperation, not as a morality play of angels and demons but as an architecture of stresses, for the first time in ten thousand years, before the roof comes down. Every family, firm and nation is broadcasting the state of its cooperation constantly, in budgets, court calendars and school curricula. The broadcast is involuntary, which is why it is reliable. Seeing it does not make us good. It makes the not-looking harder to defend, and excuses are not a small part of history.

The shiny penny is on the table. Look it up.

Peace, people.

Kari McKern is a Sydney-based independent researcher and the creator of CAMS (Complex Adaptive Model of Societies). The unified dataset, the canonical operator set and the failure log are public at neuralnations.org and the wintermute repository. Figures recomputed from the raw corpus on 31 October 2026: Figure 1 is a schematic; Figures 2 to 4 are computed readings.